

AMAL BAJPAYEE

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Education

Atal Bihari Vajpayee Indian Institute of Information Technology and Management, Gwalior **2022-2027**
Integrated B.Tech + M.Tech in Information Technology CGPA: 7.05

Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Object Oriented Programming, Machine Learning, Soft Computing, Big Data Analytics

Technical Skills

Languages: C++, Python, SQL, JSON

Backend: FastAPI, REST APIs, AsyncIO, Pydantic, OpenAPI/Swagger

AI/LLM: Prompt Engineering, Groq LLM API, LangGraph, RAG (Retrieval-Augmented Generation)

Databases: MySQL, PostgreSQL, SQLAlchemy, Alembic, FAISS, Redis

Frontend: React, Material UI, Axios

DevOps & Tools: Docker, Docker Compose, Git, GitHub, Postman, Vs Code,

Document Processing: PDF Processing, DOCX Processing **Core CS:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Experience

Lord-Tech Datus Solutions Pvt. Ltd.

Gwalior, India

Technical Project Intern (AI & Smart City)

May 2026 – Jul 2026

- Selected as a Core Technical Member for the **TIIC-funded (Rs.2,00,000)** Smart City project **NirmalGwalior**, collaborating directly with the CEO, technical team, and research mentors to build the initial product prototype.
- Designed and developed interactive web and Android application prototypes using AI-assisted development tools, translating product requirements into functional user workflows.
- Participated in system architecture, technology stack selection, and product planning discussions for a scalable AI-enabled municipal waste management platform.
- Coordinated with developers to define implementation strategy, feature roadmap, and technical execution for future production development.

Web Developer Intern

07/2025 – 12/2025

- Developed and customized client-facing web applications and dashboards based on project requirements.
- Debugged backend services and resolved API integration issues to improve application reliability and performance.
- Assisted in maintaining existing websites and implementing responsive user interfaces using modern web technologies.

Projects

AI Hiring Copilot – Agentic AI Recruitment Platform

Python, FastAPI, Groq LLM, LangGraph, RAG, FAISS, PostgreSQL, SQLAlchemy, Redis, React, Docker, REST APIs, Pydantic

- Designed and developed an Agentic AI recruitment platform that automates resume parsing, candidate evaluation, semantic job matching, interview generation, and recruiter decision support using LLMs and Retrieval-Augmented Generation (RAG).
- Developed an LLM-powered AI Hiring Copilot using Python, FastAPI, Groq LLM, React, and PostgreSQL for automated resume parsing, candidate evaluation, and recruiter workflows.
- Implemented Retrieval-Augmented Generation (RAG) using FAISS, embeddings, and semantic retrieval for intelligent resume-to-job matching and contextual AI responses.
- Designed LangGraph-based agentic workflows for resume extraction, candidate evaluation, interview question generation, and hiring recommendations.
- Built scalable FastAPI REST APIs with SQLAlchemy, Alembic, PostgreSQL, and Pydantic following a modular backend architecture.
- Optimized AI inference using Redis caching and containerized the application with Docker for production-ready deployment.

NirmalGwalior — AI-Powered Smart City C&D Waste Management Platform

[Web](#) — [App](#)

Rs.2,00,000 TIIC Funded Project — Core Technical Member

- Developed interactive web and mobile application prototypes using AI-assisted development tools to demonstrate the end-to-end Smart City waste management workflow.
- Designed user journeys, UI flows and functional screens for citizen reporting and municipal dashboard operations.
- Collaborated on system architecture, feature planning and technology stack discussions with the engineering team for future production development.

Layer Sequence Extraction of Optimized DNNs Using Side-Channel Information Leaks 05/2025 – 07/2025
Bachelor Thesis Project — Python, PyTorch, Deep Learning, LSTM, Multi-Head Attention

- Investigated side-channel information leakage techniques for architecture inference of optimized deep neural networks.
- Explored sequence modeling approaches using recurrent neural networks and attention mechanisms for reconstructing DNN layer sequences.
- Studied execution-level GPU trace patterns and operator fusion mechanisms to understand their impact on model extraction and architecture reconstruction.

Achievements

- Secured **INR 2,00,000** innovation funding under TIIC, ABV-IIITM Gwalior for the NirmalGwalior Smart City project.
- Successfully completed the **Ethical Hacking and Penetration Testing** program and virtual internship conducted by **C-DAC Noida** under the Cyber Gyan Project.

Certifications

- Deloitte Data Analytics Job Simulation – Forage (Jul 2026)